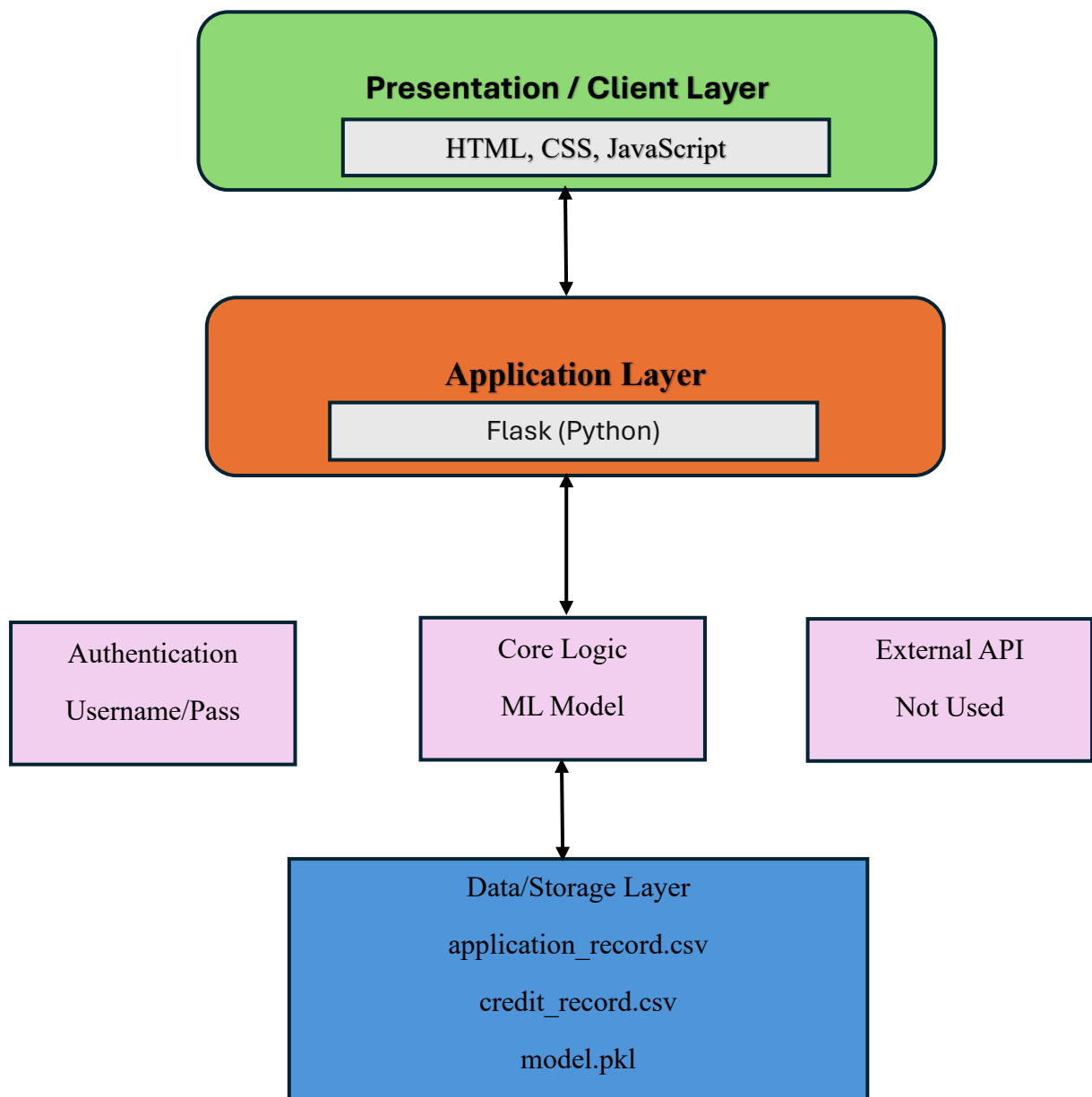


Solution Architecture

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Solution Architecture Diagram provides a high-level view of the Credit Card Approval Prediction System. It illustrates the interaction between the presentation layer, application layer, and data layer, showing how user requests are processed through the Flask application and machine learning model to generate accurate credit card approval predictions.



Component Description Table

Component Name	Description / Role in Architecture	Description / Role in Architecture
Presentation Layer	Provides the user interface for entering applicant details and displaying the prediction result.	HTML5, CSS3, JavaScript, Flask Templates
API Gateway	Receives user requests, validates input, communicates with the machine learning model, and returns prediction results.	Flask (Python Web Framework)
Core Logic Service	Performs data preprocessing, loads the trained model, and predicts whether the credit card application is approved or denied.	Python, Scikit-learn, Pandas, NumPy, Pickle
Authentication Service	Authenticates users before allowing access to the application.	Flask Session, Username & Password Authentication
Data / Storage Layer	Stores applicant datasets and the trained machine learning model used for prediction.	CSV Files, Pickle (.pkl)